Strong unextendibility of small Pauli MUB sets: exact bounds for two, three and four qubits

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Abstract

A set of mutually unbiased stabiliser bases (Pauli MUBs) is *weakly unextendible* if no further Pauli class is unbiased to all of them, and *strongly unextendible* if no vector at all is. For a set $\mathcal{S}$ in dimension d we study $\mathcal{F}(v) = \sum_e \big(|\langle e|v\rangle|^2 - |v|^2/d\big)^2$, the sum over all basis vectors e of the set; strong unextendibility is $\min_{|v|=1} \mathcal{F} > 0$. We prove exact, quantitative lower bounds. For two qubits, the minimum is exactly $1/4$. For three qubits, all 24 192 five-element sets form one Clifford orbit and satisfy $\mathcal{F} \geq (1/8 - 10^{-6})|v|^4$, while stabiliser states attain $1/8$. For four qubits, the smallest weakly unextendible sets have exactly five elements (an exhaustive search, matching Grassl's size $q^{n/2}+1$). There are 105 283 584 of them in two types. One type satisfies $\mathcal{F} \geq |v|^4/80$ exactly, so five Pauli-class bases are strongly unextendible in $d = 16$ (general, non-Pauli MUB triples are already strongly unextendible in many dimensions [5]), and its minimum (≈ 0.0524) lies below its stabiliser value $1/16$. For five qubits the smallest such sets have 8 or 9 elements (an exhaustive search excludes sizes up to 7, and one of size 9 exists). The proofs combine a twirl identity, the graph-theoretic uncertainty relation of de Gois, Hansenne and Gühne, Pauli-symmetry reduction of sum-of-squares certificates, facial reduction, and verification in exact rational arithmetic.

1 Setting

Let n qubits carry the Pauli group with symplectic space $W(2n-1, 2)$. Its points are Pauli operators up to phase, and its Lagrangians are maximal commuting classes. Disjoint Lagrangians give mutually unbiased stabiliser bases, and a weakly unextendible set is a maximal partial spread [1, 2]. Mandayam, Bandyopadhyay, Grassl and Wootters [1] constructed such sets with $d/2+1$ elements for $d = 4, 8$. They proved strong unextendibility for specific examples with a Gröbner basis computation, and noted that a general proof was open. Kantor [3] surveys maximal symplectic partial spreads and reports Grassl's sets of size $q^{n/2}+1$.

Twirl identity. For a stabiliser basis B_L, $\sum_{e\in B_L} |e\rangle\langle e|^{\otimes 2} = d^{-1}\sum_{P\in L} P^{\otimes 2}$. Hence, with C the covered Paulis and U the uncovered ones,

$$\mathcal{F}(v) = \frac{1}{d}\sum_{P\in C}\langle P\rangle^2 = \frac{1}{d}\Big(d - 1 - \sum_{P\in U}\langle P\rangle^2\Big) \qquad (|v| = 1), \tag{1}$$

since $\sum_{P\neq I}\langle P\rangle^2 = d-1$ for pure states. A stabiliser state of a Lagrangian meeting C in m points gives $\mathcal{F} = m/d$. Maximality forces $m \geq 1$, so $\min \mathcal{F} \leq m^*/d$, where m^* is the least such m.

Lovász bound. De Gois, Hansenne and Gühne [4] prove $\sum_i \langle A_i \rangle^2 \leq \vartheta(G)$ for dichotomic observables, where G is their anticommutation graph and ϑ the Lovász number [6]. Applied to U, $\min \mathcal{F} \geq (d - 1 - \vartheta)/d$. The bound is tight when $\vartheta = \alpha(G) = d - 1 - m^*$.

2 Results

Theorem 1 (two qubits)**.** *The three-element maximal partial spreads of $W(3,2)$ form one orbit, since exactly one passes through a fixed complementary pair (Witt), and for them $\min \mathcal{F} = 1/4$ exactly. Here $\vartheta = \alpha = 2$, and equivalently the six uncovered Paulis split into two pairwise-anticommuting triples.*

Theorem 2 (three qubits)**.** *The maximal partial spreads of $W(5,2)$ have sizes 5 (24 192 sets, one $\mathrm{Sp}(6,2)$-orbit) and 9 (complete sets). For every five-element set and every $v \in \mathbb{C}^8$, $\mathcal{F}(v) \geq (1/8 - 10^{-6})|v|^4$, and stabiliser states attain $1/8$. A weaker but structural bound is $\mathcal{F} \geq \frac{3-\sqrt{5}}{8}|v|^4$, from $\vartheta \leq 4 + \sqrt{5}$, certified over $\mathbb{Q}(\sqrt{5})$ in the rank-21 coherent closure of the uncovered graph.*

Method. $\mathcal{F}(v)|v|^2 - \delta|v|^6$ is written as an SOS in the degree-3 monomials and in the monomials $v_i v_j \bar{v}_k$. The Pauli group acts on both monomial spaces with centre -1, so each is a multiple of the unique 8-dimensional irrep, and invariant Gram matrices reduce to blocks of size 15 and 36. All coefficients are Gaussian rationals. At $\delta = 1/8 - \varepsilon$, a strictly interior certificate is rounded (denominator 16). Its pivot variables are restored exactly, every coefficient equation is verified in integer arithmetic, and positivity is checked by exact LDL^{T}, for $\varepsilon = 10^{-3}, \ldots, 10^{-6}$. At $\delta = 1/8$ the 160 stabiliser minimisers force the 15-block to vanish, and facial reduction leaves a 16-dimensional face with zero interior margin. This suggests that the degree-6 bound approaches $1/8$ without attaining it.

Remark (an analytic reduction). Let P_0 be the uncovered Pauli fixed by the stabiliser of the set, and let $t = \langle P_0 \rangle$. The 15 uncovered Paulis that commute with P_0 map bijectively onto $p_0^{\perp}/\langle p_0 \rangle \cong W(3,2)$, so they contribute at most 3 by two-qubit purity. The 12 that anticommute form six pairs $\{a, aP_0\}$, and they contribute exactly $(1-t^2)\sum_{j=1}^{6} |\langle \psi_+ | a_j | \psi_- \rangle|^2$. Hence $\min \mathcal{F} = 1/8$ follows from the single $4 \otimes 4$ inequality $\kappa = \max_{x,y} \sum_j |\langle x | M_j | y \rangle|^2 \leq 3$, which holds numerically with equality. Equivalently, $\| \sum_j c_j R_j \|_{\mathrm{op}}^2 \leq 3|c|^2$ for $R_j = M_j M_1^{-1}$ and all $c \in \mathbb{C}^6$, since the R_j are Hilbert–Schmidt orthogonal. Low levels of the SOS/PPT hierarchy do not certify it. The five non-identity $M_j M_1^{-1}$ anticommute along a pentagon. An exact proof of $\kappa \leq 3$ is open; it has no decomposable certificate.

Theorem 3 (four qubits)**.** *(i) $W(7,2)$ has no maximal partial spread of size 3 or 4. The smallest size is 5, and there are 105 283 584 such sets. (ii) They fall into two invariant types. For a set of the first type (lines listed in the repository), $\vartheta \leq 74/5$, so $\mathcal{F} \geq |v|^4/80$ exactly, while numerically $\min \mathcal{F} = 0.0523626 < 1/16$. A representative of the second type is a* lift *of a three-qubit five-set: for it, 120 uncovered points p project the set onto a maximal five-set of $p^{\perp}/\langle p \rangle \cong W(5,2)$, whereas the representative of the first type has no such point. It inherits a mirror partner (a second partition of the same covered set) and have numerical minimum $1/16$. (iii) For several inequivalent nine-element sets, $\min \mathcal{F} = m^*/16$ exactly ($1/16$ or $1/8$), via a facially reduced Lovász dual.*

Proposition 4 (five qubits)**.** *$W(9,2)$ has no maximal partial spread of size at most 7, and it has one of size 9 (verified independently). Hence the smallest five-qubit maximal Pauli MUB set has 8 or 9 elements; local search finds none of size 8. The data $3, 5, 5, (9)$ for $n = 2, \ldots, 5$ fit $2^{\lceil n/2 \rceil} + 1$, which is a conjecture here. The size-9 set has numerical minimum $\mathcal{F} = 1/32$, its stabiliser value.*

For $n = 6$ a coverage-biased greedy search finds a maximal partial spread of size $9 = 2^3 + 1$ (verified over all 888 832 nonsingular symmetric 6×6 matrices) and none smaller, consistent with the same pattern; this is evidence, not a proof.

Method. Fixing the X- and Z-type Lagrangians, a Lagrangian disjoint from both is the graph of a nonsingular symmetric matrix S over $\mathbb{F}_2$, and two graphs are disjoint iff $S + S'$ is nonsingular. A maximal partial spread is thus a set of nonsingular symmetric matrices with nonsingular pairwise differences that leaves no nonsingular T with all $T + S_i$ nonsingular. For $n = 5$ there are 13 888 such matrices; $S_1 = I$ without loss of generality, and S_2 is reduced modulo the orthogonal group of order 720, followed by an exhaustive depth-first search with a valid covering prune for size 7. The same code reproduces the counts 56 ($n = 3$) and 896 ($n = 4$) found by direct enumeration.
Method for (i). By Witt's theorem, $\mathrm{Sp}(8, 2)$ is transitive on ordered pairs of complementary Lagrangians, so every candidate can be taken to contain a fixed pair. The remaining one or two Lagrangians are enumerated exhaustively. As a control, the same code finds no sizes 3 and 4 for three qubits, and finds 56 five-sets through a fixed pair, matching $24\,192 \cdot 20/(135 \cdot 64) = 56$. *Method for (iii).* The maximum independent sets of the uncovered graph are the Lagrangians meeting C in m^* points, minus those points. Complementary slackness with them gives linear equations on the dual. Imposing these equations makes rounding succeed, and the certificate is checked by exact LDL^{T}.

Remark (two qutrits). For the 432 five-element maximal partial spreads of $W(3, 3)$, $\mathcal{F} \geq |v|^4/9$ exactly, via an S_6/Jucys–Murphy identity. The true minimum is numerically $(207 - 19\sqrt{57})/432 \approx 0.14711$, which is an identification, not a proof. Mirror partners there lie in opposite orbits and are exchanged only by antiunitaries; a third-moment witness of size 16/27 separates them. In characteristic 2 the corresponding partners are Clifford-equivalent.

3 Summary and open questions

d	smallest size	$\min \mathcal{F}$	status
4	3	$1/4$	exact
8	5	$\in [1/8 - 10^{-6}, 1/8]$	exact interval
9	5	$\geq 1/9$; ≈ 0.14711	exact bound; numerical value
16	5	$\geq 1/80$; ≈ 0.05236 (one type)	exact bound; numerical value
32	8 or 9	$\approx 1/32$ (size-9 set)	sizes exact; value numerical

Open: an exact certificate at 1/8 (degree 8, or a structural proof); closed forms for the non-stabiliser optima; the strong unextendibility constant for all five-element sets in $d = 16$ and beyond. *Scope*: all statements concern *Pauli-class* MUB sets; strongly unextendible sets of only three (non-Pauli) MUBs exist in many dimensions [5]. *Not claimed*: novelty of the weak-unextendibility census at $d = 16$, which may already be in Grassl's tables [3]. All certificates are in the repository (`data/*.json`) with the scripts that verify them.

References

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